



**SYDNEY BOYS HIGH  
SCHOOL**  
MOORE PARK, SURRY HILLS

**2015**  
**HIGHER SCHOOL CERTIFICATE**  
**ASSESSMENT TASK #2**

# Mathematics

## General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
- Board approved calculators may be used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

## Total Marks – 69

- Attempt questions 1 – 12.
- All questions are not of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

### Section A (24 Marks)

#### Questions 1 to 7. (7 marks)

Indicate which of the answers A, B, C, or D is the correct answer.

Marks

**Write the answer on the separate answer sheet.**

- (1) The gradient of the tangent to the curve  $y = (x - 1)(x^2 + 1)$  at the point where  $x = \frac{1}{2}$  is: 1

A:  $-\frac{4}{3}$

B:  $\frac{3}{4}$

C:  $\frac{4}{3}$

D:  $-\frac{3}{4}$

- (2) For what values of  $x$  is the curve  $f(x) = 2x^2 - x^3$  increasing? 1

A:  $x > \frac{2}{3}$

B:  $x < 0, x > \frac{4}{3}$

C:  $x < \frac{2}{3}$

D:  $0 < x < \frac{4}{3}$

- (3) What is the slope of the line containing the points  $(-9, 2)$  and  $(3, 14)$ . 1

A:  $\frac{4}{3}$

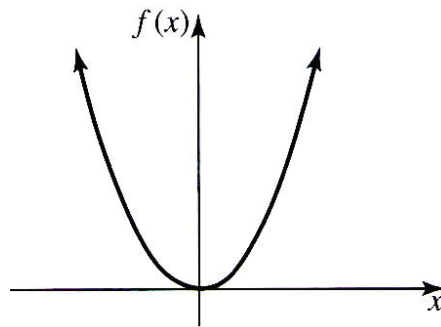
B:  $-\frac{1}{2}$

C: 1

D: -2

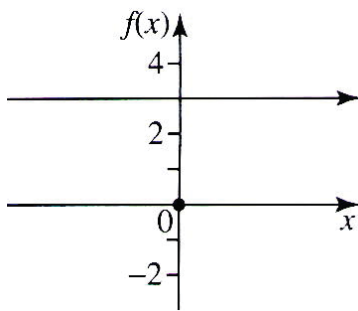
(4) The figure below is the graph of  $f(x) = 3x^2$ :

1

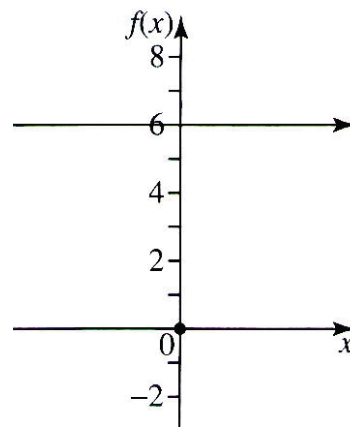


Which of the following represents the graph of the second derivative?

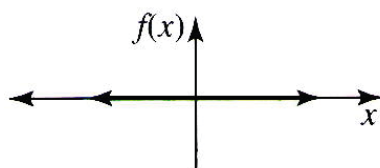
A:



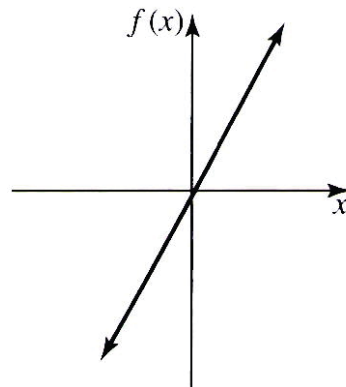
B:



C:



D:



(5) Differentiate the following equation:  $y = 6x^{-3}$

1

**A:**  $\frac{dy}{dx} = -18x^{-2}$

**B:**  $\frac{dy}{dx} = -12x^{-3}$

**C:**  $\frac{dy}{dx} = -18x^{-4}$

**D:**  $\frac{dy}{dx} = -3x^{-2}$

(6) It is known that  $f''(a) = 0$ . The point  $(a, f(a))$  is:

**A:** a minimum turning point

**B:** a maximum turning point

**C:** a horizontal point of inflection

**D:** not determined (insufficient information)

(7) The graph with equation  $y = x^2$  is translated 3 units down and 2 units to the right. Which equation represents the resulting graph?

**A:**  $y = (x - 2)^2 + 3$

**B:**  $y = (x - 2)^2 - 3$

**C:**  $y = (x + 2)^2 + 3$

**D:**  $y = (x + 2)^2 - 3$

**Question 8** (17 marks) (Start a new booklet)

**Marks**

(a) Differentiate the following:

(i)  $\ln x^3$  1

(ii)  $e^{2x+1}$  1

(iii)  $e^x \ln x$  1

(b) Find

(i)  $\int (3x^2 - 4x + 1) dx$  2

(ii)  $\int \left( \frac{1}{x} - \frac{1}{x-1} \right) dx$  2

(iii)  $\int \left( \frac{1}{e^{2x}} \right) dx$  2

(c) Evaluate

(i)  $\int_{-1}^3 (2x - 1) dx$  2

(i)  $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left( \frac{\cos x}{1 + \sin x} \right) dx$  2

(d) (i) Copy and complete the table for  $f(x) = e^{x^2}$  correct to 4 decimal places. 2

|        |   |     |   |     |   |
|--------|---|-----|---|-----|---|
| $x$    | 0 | 0.5 | 1 | 1.5 | 2 |
| $f(x)$ | 1 |     |   |     |   |

(ii) Use Simpson's Rule with the above 5 function values to find an approximation to  $\int_0^2 e^{x^2} dx$  correct to 3 decimal places. 2

## Section B (23 Marks)

START A NEW BOOKLET

### Question 9 (12 Marks)

Marks

(a) In an  $x$ - $y$  plane in your answer booklet:

- (i) Sketch the line through  $C(-3,0)$  which makes an angle of  $45^\circ$  with the positive direction of the  $x$ -axis. Also sketch the line  $x + y = 4$ , which meets the first line at  $A$ , and the  $x$ -axis at  $B$ . 1
- (ii) Show that  $AC$  is perpendicular to  $AB$ . 1
- (iii) Find the equation of the line through  $B$ , which is parallel to  $AC$ . 2
- (iv) Show that the equation of the line through the point  $C$  parallel to the line  $AB$  is  $x + y + 3 = 0$ . 1
- (v) The lines from parts (iii) and (iv) meet at  $D$ . Find the coordinates of  $D$ . 2
- (vi) What type of quadrilateral is  $ABDC$ ? Give reasons. 2

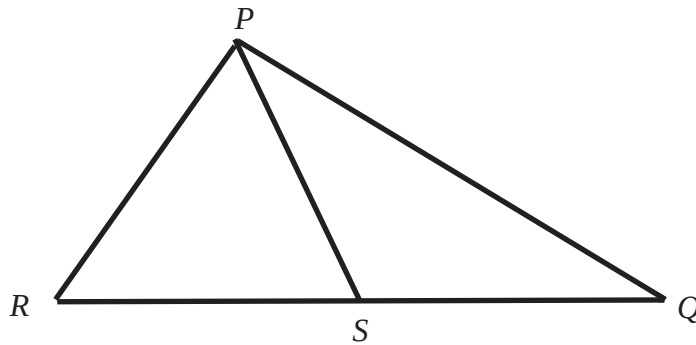
(b) A car's velocity  $v$  in metres per second is recorded each second as it accelerates along a drag strip. The table below gives the results. 3

|                              |   |    |    |    |    |    |
|------------------------------|---|----|----|----|----|----|
| $t \text{ (s)}$              | 0 | 1  | 2  | 3  | 4  | 5  |
| $v \text{ (ms}^{-1}\text{)}$ | 0 | 15 | 31 | 48 | 64 | 83 |

Given that the distance travelled may be found by calculating the area under a velocity/time graph, use the trapezoidal rule to estimate the distance travelled by the car in the first five seconds.

**Question 10** (11 Marks)

(a)



In the diagram  $\angle QPR = 90^\circ$ ,  $PS = SQ$ .

(i) Copy the diagram to your answer booklet.

(ii) Construct  $ST \perp PQ$  (Sketch only).

**1**

(ii) Prove that  $PS = \frac{1}{2} RQ$ .

**3**

(b) A triangle has vertices  $A(4, 0)$ ,  $B(-4, 0)$  and  $C(0, 6)$ .

(i) Draw a neat sketch of the triangle in your answer booklet.

**1**

(ii) State the coordinates of  $D$  and  $E$ , the mid-points of  $CA$  and  $CB$  respectively.

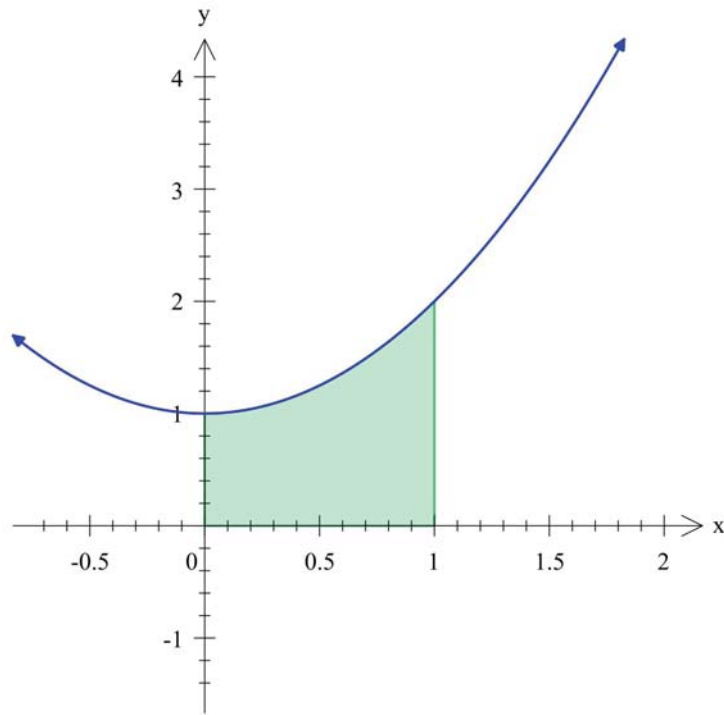
**1**

(iii) Show that the medians  $BD$  and  $EA$  meet on the  $y$ -axis.

**3**

(c)

2



In the diagram the shaded region is bounded by the parabola  $y = x^2 + 1$ , the  $x$ -axis and the lines  $x = 0$ , and  $x = 1$ .  
Find the volume of the solid formed when the shaded region is rotated about the  $x$ -axis.

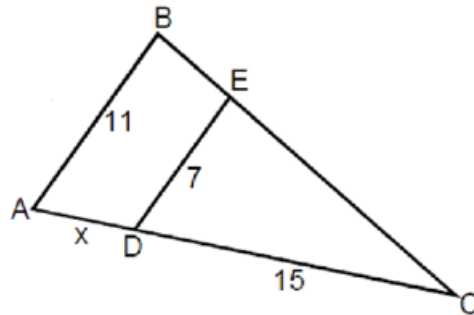
**Section C (22 Marks)**

START A NEW BOOKLET

**Question 11 (11 Marks)**

(a)

3



In the diagram above,  $AB \parallel DE$ .

Find the value of  $x$ , correct to 2 decimal places, giving full reasons.

(b) Consider the curve with equation  $y = 2x^3 - 9x^2 + 12x - 3$ .

- |       |                                                                                                   |   |
|-------|---------------------------------------------------------------------------------------------------|---|
| (i)   | Find the co-ordinates of the stationary points and determine their nature.                        | 4 |
| (ii)  | Find the co-ordinates of any points of inflexion.                                                 | 2 |
| (iii) | Sketch the curve for the domain $0 \leq x \leq 3$ . (Do not attempt to find any $x$ -intercepts.) | 2 |

**Question 12 (11 Marks)**

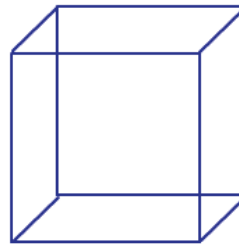
- (a) For a certain curve  $y = f(x)$ ,  $f''(x) = 6x - 1$ . 2

Find the equation of the curve given that it passes through the point (1, -2) with gradient -3.

- (b) (i) Differentiate  $e^x \sqrt{x}$ . 2

- (ii) Hence evaluate  $\int_1^2 \frac{e^x(1+2x)}{\sqrt{x}} dx$ . 2

- (c) A sealed tin rectangular box is to have a square base and a volume of  $64 \text{ cm}^3$ . If the length of the edge of the base is  $x \text{ cm}$ :



- (i) Express the height of the box in terms of  $x$ . 1
- (ii) Show that the total surface area  $y \text{ cm}^2$  of the box is given by 2  
$$y = \frac{256}{x} + 2x^2.$$
- (iii) Find the minimum surface area of the box, and its dimensions. 2

***This is the end of the paper.***



## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, x > 0$



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**2015**

HSC Task #2

# Mathematics

## Suggested Solutions & Markers' Comments

| QUESTION | Marker     |
|----------|------------|
| 1 – 7    | –          |
| 8        | VL see PSP |
| 9 & 10   | JC         |
| 11 & 12  | RB         |

### Multiple Choice Answers

1. B
2. D
3. C
4. B
5. C
6. D
7. B

Q.1  $y' = u'v + uv'$  where  $u = x-1$   
 $= x^2+1 + (x-1) \times 2x$   $u' = 1$   
 $= x^2+1 + 2x^2-2x$   $v = x^2+1$   
 $= 3x^2-2x+1$   $v' = 2x$

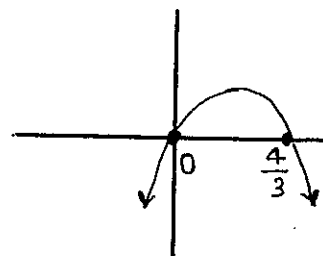
$$f'(\frac{1}{2}) = 3(\frac{1}{2})^2 - 2(\frac{1}{2}) + 1$$

$$= \frac{3}{4} - 1 + 1$$

$$= \frac{3}{4}$$

B

Q.2  $f'(x) = 4x - 3x^2$   
 curve increases when  $f'(x) > 0$   
 $4x - 3x^2 > 0$   
 $x(4-3x) > 0$   
 $\therefore 0 < x < \frac{4}{3}$



D

Q.3 slope =  $\frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{14 - 2}{3 - (-9)}$$

$$= 1$$

C

Q. 4

$$f(x) = 3x^2$$

$$f'(x) = 6x$$

$$f''(x) = 6$$

B

Q. 5

$$y = 6x^{-3}$$

$$\frac{dy}{dx} = -18x^{-4}$$

C

Q. 6

D

Q. 7

$$y = (x - 2)^2 - 3$$

↑  
translated 2 units  
to the right

← translated  
3 units down

B

### Question 8

$$(a) (i) \quad \frac{d}{dx} \ln x^3 = \frac{d}{dx} 3 \ln x \\ = \frac{3}{x}$$

a)(i) Those who wrote the following received no marks because they are all incorrect:

$$(3x^2)^{-1}, \quad \frac{1}{x^3}, \quad \frac{x}{3}, \quad \frac{3 \ln x^2}{\ln x^3}, \quad \frac{2x}{x^3}, \quad 3 \ln x^2, \\ 3x^3, \quad 3e^x, \quad \frac{3}{x^2}, \quad 3 \ln e^x, \quad 3x^2 \ln x^3$$

Those who wrote  $\frac{3x^2}{x^3}$  received  $\frac{1}{2}$  out of 1 mark because they should have simplified their answer to  $\frac{3}{x}$ .

---

$$(ii) \quad \frac{d}{dx} e^{2x+1} = 2e^{2x+1}$$

---

a)(ii) Those who wrote the following received no marks because they were all incorrect:

$$2e^{2x}, e^2, e^{2x+1}, 2xe^{2x+1}, (2x+1)e^{2x+1}, (2x+1)e^{2x}, \\ \frac{1}{2}e^{2x+1}$$

---

$$\begin{aligned}
 \text{(iii)} \quad \frac{d}{dx} e^x \ln x &= u'v + uv' \quad \text{where } u = e^x \\
 &= e^x \ln x + \frac{e^x}{x} \quad u' = e^x \\
 &= e^x \left( \ln x + \frac{1}{x} \right) \quad v = \ln x \\
 &\quad v' = \frac{1}{x}
 \end{aligned}$$


---

a)(iii) The product rule should be used.

Those who did not use the product rule wrote the following which were all incorrect:

$$e^x \times \frac{1}{x} = \frac{e^x}{x}, \quad e^x \ln x, \quad \frac{1}{x^x}, \quad x^{-e^x}, \quad e^x, \quad 2xe^x \ln x$$


---

$$\text{(b) (i)} \quad \int (3x^2 - 4x + 1) dx = x^3 - 2x^2 + x + C$$

$$\text{(ii)} \quad \int \left( \frac{1}{x} - \frac{1}{x-1} \right) dx = \ln x - \ln(x-1) + C$$


---

b)  $\rightarrow$  1 mark was lost if the constant was not included.

b)(i) well answered overall

b)(ii) The following scored zero marks:

$$\begin{aligned}
 &-1 - (x-1)^{-1} + C, \quad \ln x + \ln(x-1) + C, \quad 1 + C, \quad -x + C, \\
 &\frac{-2}{x-1}, \quad \frac{x^0}{0} = \text{undefined}, \quad \frac{\ln(x)}{\ln(x-1)}, \quad \frac{1}{x} + C, \quad \ln x - \ln x - \ln x
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \int \left( \frac{1}{e^{2x}} \right) dx &= \int e^{-2x} dx \\
 &= \frac{e^{-2x}}{-2} + C
 \end{aligned}$$

(b) (iii) If the negative sign was omitted from the first term, one mark was lost  
i.e.  $\frac{e^{-2x}}{-2} + c$  is the correct answer (not  $\frac{e^{-2x}}{2} + c$ )

This question was not answered correctly overall with incorrect answers such as:

$$\frac{\ln e^{2x}}{2e^{2x}}, \ln e^{2x} + c, e^{-2x} + c, 2x + c, \frac{e^{2x-1}}{2x},$$

$$\frac{x}{e^{2x}} + c, \frac{1}{e^x} + c, \frac{1}{2} \ln(e^{2x}), -3e^{-3x} + c,$$

$$\frac{e^{-2x+1}}{-2x+1} + c, \ln e^{2x} \cdot \frac{1}{2e^{2x}}, \frac{1}{e^{2x^2}} + c$$

---

$$\begin{aligned} \text{(c) (i)} \quad \int_{-1}^3 (2x-1) dx &= \left[ x^2 - x \right]_{-1}^3 \\ &= (3^2 - 3) - ((-1)^2 - (-1)) \\ &= (9 - 3) - (1 + 1) \\ &= 4 \end{aligned}$$

$$(c) (i) \quad [x^2 - x]_{-1}^3$$

$$= (3^2 - 3) - ((-1)^2 - (-1))$$

$$= (9 - 3) - (1 + 1)$$

$$= 4$$

If grouping symbols were not used, students incorrectly wrote  $6 - (-2) = 8$

$$\left\{ \begin{array}{l} \text{or incorrectly wrote} \\ 6 - 1 + 1 \\ = 6 \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{or incorrectly wrote} \\ 6 - (1 - 1) \\ = 6 \end{array} \right.$$

The negative sign on the lower limit was incorrectly left out (1 was used instead of -1)

Most students answered this correctly because grouping symbols were used.

$$(c) (ii) \quad \int_{\pi/6}^{\pi/3} \left( \frac{\cos x}{1 + \sin x} \right) dx = \left[ \ln(1 + \sin x) \right]_{\pi/6}^{\pi/3}$$

$$= \ln\left(1 + \frac{\sqrt{3}}{2}\right) - \ln\left(1 + \frac{1}{2}\right)$$

$$= \ln\left(\frac{2 + \sqrt{3}}{2}\right) - \ln\left(\frac{3}{2}\right)$$

$$= \ln\left(\frac{\frac{2 + \sqrt{3}}{2}}{\frac{3}{2}}\right)$$

$$= \ln\left(\frac{2 + \sqrt{3}}{3}\right)$$

(d) (i)

|        |   |        |        |        |         |
|--------|---|--------|--------|--------|---------|
| $x$    | 0 | 0.5    | 1      | 1.5    | 2       |
| $f(x)$ | 1 | 1.2840 | 2.7183 | 9.4877 | 54.5982 |

$$\begin{aligned} \text{(ii)} \quad \int_0^2 e^{x^2} dx &\approx \frac{h}{3} \left[ f(0) + f(2) + 4(f(0.5) + f(1.5)) \right. \\ &\quad \left. + 2(f(1)) \right] \\ &= \frac{\frac{2-0}{4}}{3} \left[ 1 + 54.5982 + 4(1.2840 + 9.4877) \right. \\ &\quad \left. + 2(2.7183) \right] \\ &= 17.354 \text{ (correct to 3 decimal places)} \end{aligned}$$

(d) (i) Some students did not round to 4 decimal places (instead they rounded to 3 decimal places and lost 1 mark).

When completing the table, approximations were not written and instead exact values  $e^{\frac{1}{4}}$ ,  $e^1$ ,  $e^{2\frac{1}{4}}$ ,  $e^4$  were written. No marks were awarded for exact values.

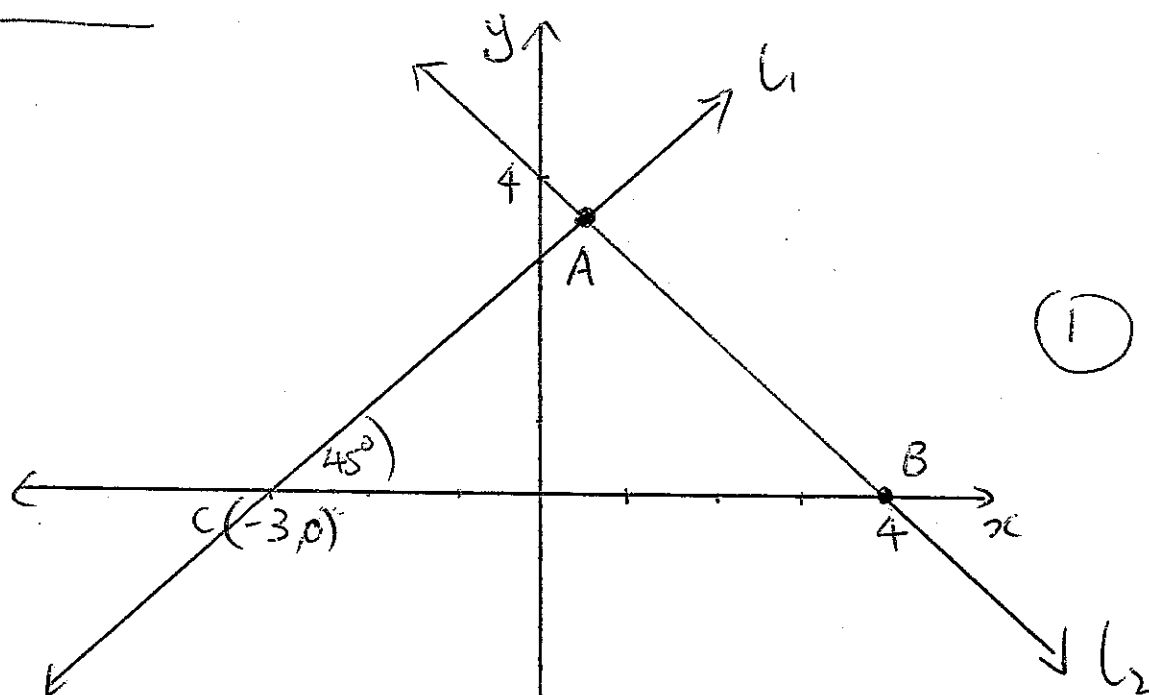
$\frac{1}{2}$  mark was lost for each incorrect approximation.

(d)(ii) Overall, Simpson's Rule was written and applied correctly. In a few cases,  $\frac{h}{3}$  was incorrectly calculated as  $\frac{2-0}{4} = \frac{1}{2}$  (instead of  $\frac{1}{6}$ )  $\frac{1}{2}$  mark was lost if  $\frac{h}{3}$  was incorrect.

If Simpson's Rule was written and applied correctly however the final answer was not rounded to 3 decimal places or incorrect,  $\frac{1}{2}$  mark was lost.

# Question 9

a) (ii)



Almost all the students were able to attain full marks

$$(ii) \quad m_{L_1} = \tan 45^\circ$$

$$m_{L_1} = 1$$

$$y = -x + 4$$

$$m_{L_2} = -1$$

$$m_{L_1} \times m_{L_2} = 1 \times -1 = -1 \quad (1)$$

$$\therefore AC \perp AB$$

Almost all the students were able to attain full marks

$$(iii) \quad y - 0 = 1(x - 4) \quad (1)$$

$$\boxed{y = x - 4}$$

$$(1)$$

Almost all the students were able to attain full marks

$$(iv) \quad y - 0 = -1(x + 3)$$

$$y = -x - 3$$

$$x + y + 3 = 0$$

$$(1)$$

Almost all the students were able to attain full marks

Question 9 (continued)

(2)

a) (v)  $y = x - 4$  — (1)

$x + y + 3 = 0$  — (2)

Sub (1) in (2),

$$x + x - 4 + 3 = 0$$

$$2x - 1 = 0$$

$$\boxed{x = \frac{1}{2}} \quad (1)$$

$$y = \frac{1}{2} - 4$$

$$\boxed{y = -3\frac{1}{2}} \quad (1)$$

$$\boxed{\therefore D = \left(\frac{1}{2}, -3\frac{1}{2}\right)}$$

Almost all the students were able to attain full marks

(vi) ABCD is a square (1)

\* All sides equal, adjacent sides are perpendicular. (1)

This part was poorly done, most students weren't able to identify the type of quadrilateral

Question 9 (continued)

(3)

$$\begin{aligned} \text{b). } d &= \frac{h}{2} [f(a) + 2f(x_n) + f(b)] & (1) \\ &= \frac{1}{2} [0 + 2(15 + 31 + 48 + 64) + 83] & (1) \end{aligned}$$

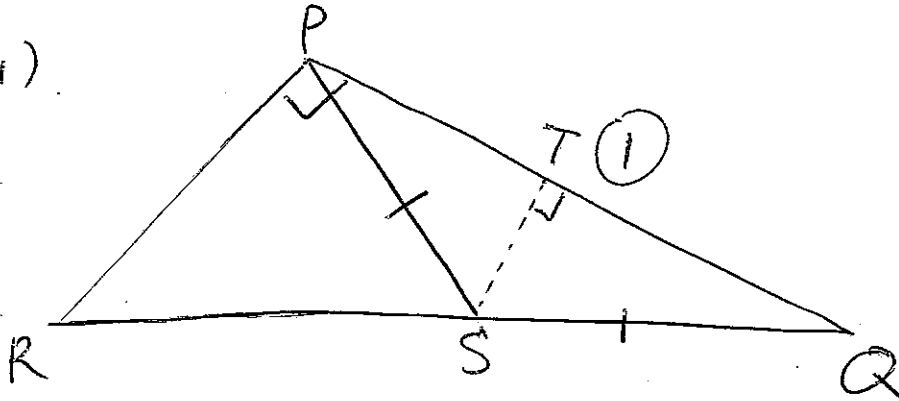
$$d = 199 \frac{1}{2} \text{ m} \quad (1)$$

Most students were able to attain full marks

# Question 10 :

(4)

a) (i).



Almost all the students were able to attain full marks

(iii). Let  $\angle SPT = x$

$\angle SQT = \angle SPT = x$  (base  $\angle$ s of isos.  $\Delta$ )

$\angle RPS = 90^\circ - x$  (comp.  $\angle$ )

$\angle PRQ = 180^\circ - 90^\circ - x$  ( $\angle$ s sum of a right- $\angle$   $\Delta$ ) (1)  
 $= 90^\circ - x$

$\therefore \angle PRQ = \angle RPS = 90^\circ - x$

$\therefore \Delta RSP$  is isosceles with  $PS = RS$  (1)

$\therefore PS = RS = SQ$

$\therefore PS = \frac{1}{2}(RS + SQ)$  (1)

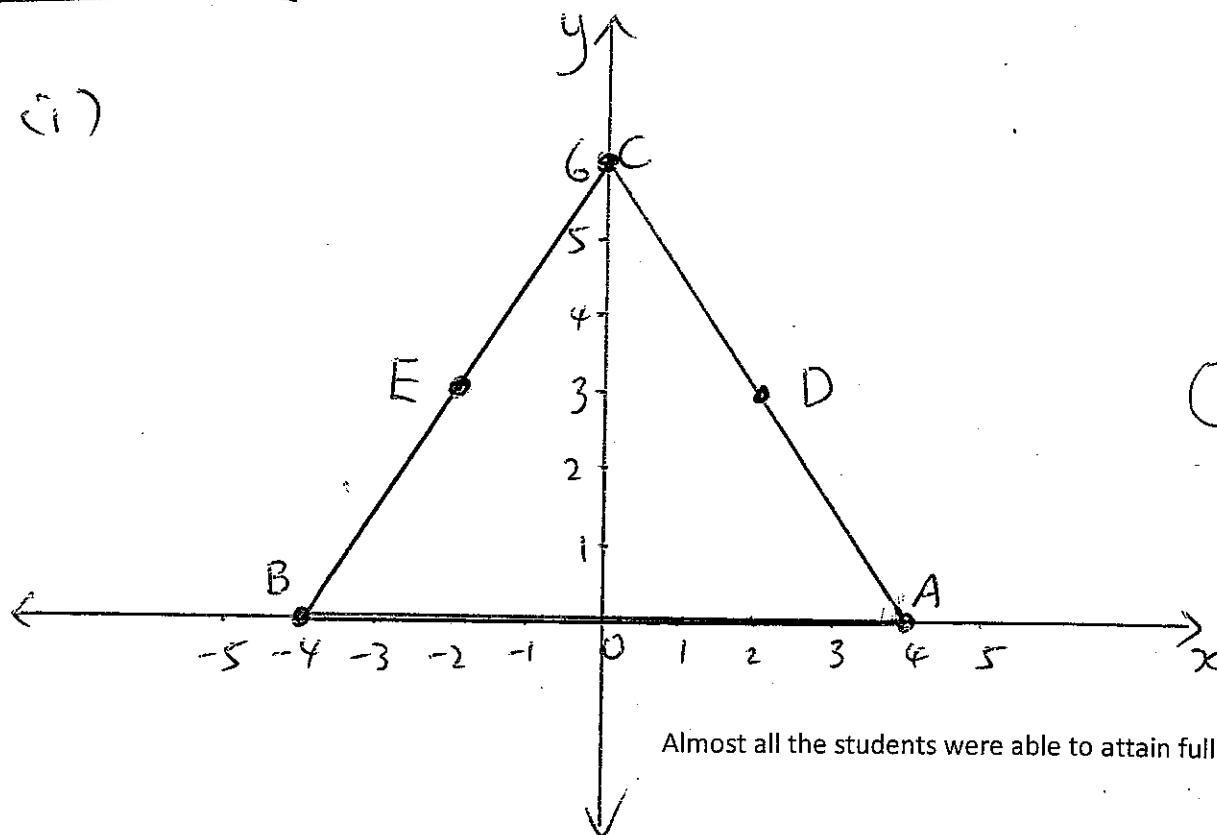
$PS = \frac{1}{2} RQ$  //

Most students were able to write out a logical proof for this question

# Question 10 (continued)

(5)

b). (i)



(1)

Almost all the students were able to attain full marks

(ii)  $D(2, 3)$   $\left(\frac{1}{2}\right)$

Almost all the students were able to attain full marks

$E(-2, 3)$   $\left(\frac{1}{2}\right)$

(iii)  $m_{BD} = \frac{3-0}{2+4}$   
 $m_{BD} = \frac{1}{2}$

$m_{AE} = \frac{3-0}{-2-4}$   
 $= -\frac{1}{2}$

(1)

Equation of BD:  $y-0 = \frac{1}{2}(x+4)$

$y = \frac{1}{2}x + 2$  — (1)

Equation of AE:  $y-0 = -\frac{1}{2}(x-4)$

$y = -\frac{1}{2}x + 2$  — (2)

Equate (1) & (2)

$\frac{1}{2}x + 2 = -\frac{1}{2}x + 2$   
 $\boxed{x=0}$

$y = \frac{1}{2}(0) + 2$   
 $\boxed{y=2}$  (1)

$\therefore$  Medians meet at  $(0, 2)$  which is on the y-axis.

Question 10 (continued)

(6)

c)  $V = \pi \int_a^b y^2 \cdot dx$

$$= \pi \int_0^1 (x^2 + 1)^2 \cdot dx$$

$$= \pi \int_0^1 x^4 + 2x^2 + 1 \cdot dx$$

(1)

$$= \pi \left[ \frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1$$

$$= \pi \left[ \frac{1}{5} + \frac{2}{3} + 1 - 0 \right]$$

$$\therefore V = \frac{28\pi}{15}$$

(1)

Deduct  $\left(\frac{1}{2}\right)$  for not including  $\pi$ .

Almost all the students were able to attain full marks

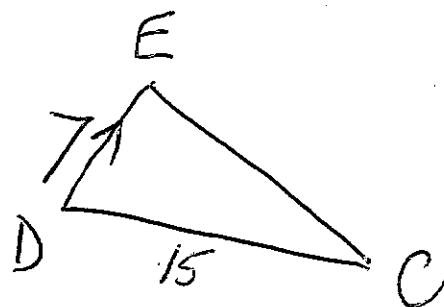
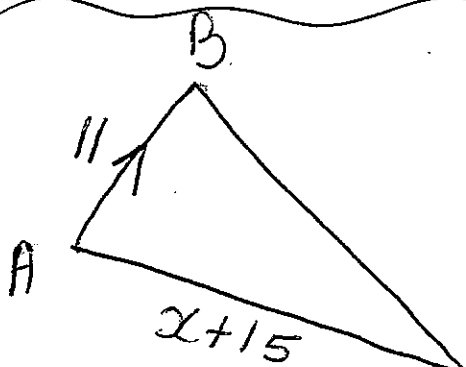
# Assessment Task 2 Unit 2015

## March/April Section C

11

11

a)



In  $\triangle ABC$  and  $\triangle DEC$ ,

$\hat{C}$  is common

$\hat{BAC} = \hat{EDC}$  angles in the corresponding position as  $AB \parallel DE$  given

$\therefore \triangle ABC \parallel \triangle DEC$  2 angle test. ①

thus  $\frac{AB}{DE} = \frac{AC}{DC} = \frac{BC}{EC}$  ①

$$\frac{11}{7} = \frac{x+15}{15}$$

$$11 \times 15 = 7(x+15)$$

$$165 = 7x + 105$$

$$7x = 60$$

$$x = \frac{60}{7} \approx 8.57 \text{ 2DP } ①$$

or stating the proportional division theorem  
"A line parallel to 1 side of a triangle divides the other sides in the same ratio"

\* Well answered but students did get the ratio of sides wrong.

11 (b)  $y = 2x^3 - 9x^2 + 12x - 3$  Well answered  
 $y' = 6x^2 - 18x + 12$  but Max/min has  
 $y'' = 12x - 18$  to be established.  
 Sign change for  
 inflexion established.

(i) Stat points exist when  $y' = 0$  Graph domain  
 $0 \leq x \leq 3$   
 $6x^2 - 18x + 12 = 0$  only and  
 $\div 6$   $x^2 - 3x + 2 = 0$  y values then.  
 $(x-2)(x-1) = 0$   
 $x=1, x=2$

At  $x=1$ ,  $y = 2 - 9 + 12 - 3 = 2$  (1, 2) ①

At  $x=2$ ,  $y = 16 - 36 + 24 - 3 = 1$  (2, 1) ①

At (1, 2)  $y'' = 12 - 18 = -6 < 0$  MAX st pt. ①

At (2, 1)  $y'' = 24 - 18 = 6 > 0$  min st pt. ①

(ii) Inflexions occur when  $y'' = 0$  and there is a  
 sign change.

$y'' = 12x - 18 = 0$   
 $12x = 18$   
 $x = 1\frac{1}{2}$  ①

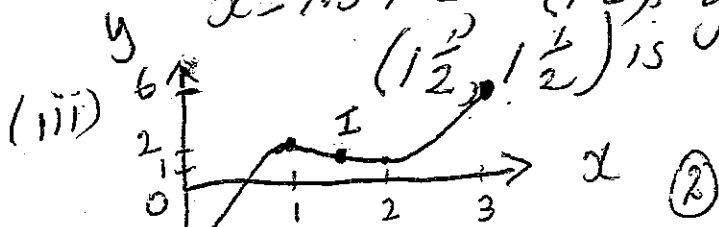
At  $x = 1\frac{1}{2}$   $y = 6.75 - 20.25 + 18 - 3 = 1\frac{1}{2}$  (1.5, 1.5) ①

Sign change?

At  $x = 1.5 - \epsilon$ , (1.4)  $y'' = 12 \times 1.4 - 18 = -1.2 < 0$  ①

$x = 1.5 + \epsilon$ , (1.6)  $y'' = 12 \times 1.6 - 18 = 1.2 > 0$  ①

(1.5, 1.5) is a pt of inflexion



11

12 (a)  $f''(x) = 6x - 1$

$$f'(x) = \int (6x - 1) dx$$

$$f'(x) = \frac{6x^2}{2} - x + C_1$$

data

$x=1, \quad -3 = 3 - 1 + C_1$   
 $f'(x) = -3 \quad C_1 = -5$

$$f'(x) = 3x^2 - x - 5 \quad (1)$$

$$f(x) = \int (3x^2 - x - 5) dx$$

$$= \frac{3x^3}{3} - \frac{x^2}{2} - 5x + C_2$$

$$f(x) = x^3 - \frac{x^2}{2} - 5x + C_2$$

data  $(1, -2) \quad -2 = 1 - \frac{1}{2} - 5 + C_2$

$$C_2 = 2\frac{1}{2}$$

$$\therefore f(x) = x^3 - \frac{x^2}{2} - 5x + 2\frac{1}{2} \quad (i)$$

Often made mistakes,

: found  $f'(x)$  but not the first constant.

: many found  $f(x)$  but not the 2<sup>nd</sup> constant.

: many students left out the question data completely.

$$12 \text{ (b) (i) } \frac{d}{dx} (e^x \cdot x^{\frac{1}{2}})$$

$$= e^x \times \frac{1}{2} x^{-\frac{1}{2}} + x^{\frac{1}{2}} \cdot e^x$$

$$= \frac{e^x}{2\sqrt{x}} + e^x \cdot \sqrt{x} = \frac{e^x + e^x \cdot 2x}{2\sqrt{x}}$$

Most students handled the product rule well. Many thought I did not simplify enough to help in part (ii)

$$= \frac{e^x(1+2x)}{2\sqrt{x}} \quad \text{not rationalised. (far enough)}$$

$$(ii) \quad (2) \int_1^2 \frac{e^x(1+2x)}{(2)\sqrt{x}} dx = (2e^x \sqrt{x}) \Big|_1^2$$

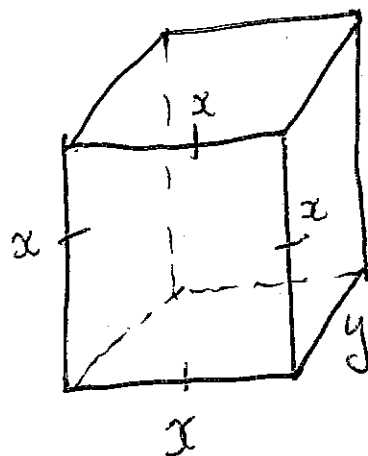
$$= 2(e^2 \cdot \sqrt{2} - e^1 \cdot \sqrt{1})$$

$$= 2(\sqrt{2}e^2 - e^1)$$

$$= 2e(\sqrt{2}e - 1) \quad \text{exact. } (2)$$

Generally well answered but the  $2\int$  was often missing.

12 (c)



(i)  $V = x \times x \times y$   $y = \text{height}$

$$64 = x^2 y$$

$$y = \frac{64}{x^2}$$

height  $y = \frac{64}{x^2}$

① easily found

(ii) total S.A. =  $x^2 + x^2 + xy + xy + xy + xy$   
 $= 2x^2 + 4xy$   
 $= 2x^2 + 4x \times \frac{64}{x^2}$

$$y = \frac{256}{x} + 2x^2$$

② well answered -

(iii)  $y = 256x^{-1} + 2x^2$

$$y' = -256x^{-2} + 4x$$

$$y'' = 512x^{-3} + 4$$

When  $y' = 0$   $-\frac{256}{x^2} + 4x = 0$

obviously  $x \neq 0$

$$\frac{-256 + 4x^3}{x^2} = 0$$

$$4x^3 = 256$$

$$x^3 = 64$$

$$x = 4 \text{ cm}$$

If dimensions are  $4 \times 4 \times 4$

$$\begin{aligned} \text{then } y'' &= \frac{512}{3} + 4 \\ &= \frac{4}{8} + 4 = 12 \\ &\Rightarrow \text{min value} \end{aligned}$$

$\uparrow$   
 $\frac{64}{4^2}$

So minimum S.A is when dimensions are  $4 \times 4 \times 4$  cm and its value

$$\begin{aligned} \text{SA is } 2 \times 4^2 + \frac{256}{4} \\ &= 32 + 64 \\ &= 96 \text{ cm}^2 \quad (2) \end{aligned}$$

a minimum was often not found by either using  $y'$  or  $y''$ .  
Dimensions often not finally quoted  
the minimum surface area often not found as a conclusion.